$\boxed{4.10}$ HW # 11

To be determined: R_{30}, R_{31} & R_{32}

Note: $R_{nl}(r) = \frac{1}{r}\rho^{l+1} e^{-\rho} V(\rho)$

where $V(\rho)$ is a polynomial of degree $j_{max} = n-l-1$.

Use $\rightarrow$ $c_{j+1} = \frac{2(j+l+1-n)}{(j+1)(j+2l+2)} c_j$ to find $V(\rho)$

$R_{30} \rightarrow j_{max} = 3-1=2 \rightarrow c_{j+1} = \frac{2(j-2)\ c_j}{(j+1)(j+2)}$

$$c_1 = \frac{-4}{2} c_0 = -2c_0$$

$$c_2 = \frac{-2\cdot(\overbrace{-2c_0}^{c_1})}{2\times 3} = \frac{2c_0}{3}$$

$$\Rightarrow R_{30} = \frac{1}{r}\rho^{l+1} e^{-\rho} c_0 \left[1 - 2\rho + \frac{2}{3}\rho^2\right]; \quad \boxed{\rho \equiv \frac{r}{3a}}$$

$$= \frac{1}{r3a} e^{-r/3a} c_0 \left[1 - \frac{2r}{3a_0} + \frac{2r^2}{27a^2}\right]$$

$R_{31} \rightarrow j_{max} = 3-1-1 = 1 \rightarrow c_{j+1} = \frac{2(j-1)}{(j+1)(j+4)} c_j$

$$c_1 = \frac{-2}{4} c_0$$

$$\Rightarrow R_{31} = \frac{1}{r}\rho^2 e^{-\rho} c_0 \left[1 + \frac{1}{2}\rho^2\right]; \quad \rho = \frac{r}{3a}$$

$$= c_0 \frac{r}{9a^2} e^{-r/3a}\left[1 - \frac{r}{6a}\right]$$

$R_{32} \rightarrow j_{max} = 3-2-1 = 0 \Rightarrow$ only c_0 with $\rho = r/3a$

$$\Rightarrow R_{32} = \frac{c_0}{r}\rho^3 e^{-\rho} = \frac{c_0 r^2}{27a^3} e^{-r/3a}.$$

4

$$\Rightarrow \langle x^2 \rangle = \frac{1}{\pi a}\frac{1}{(8a^2)^2}\frac{16\pi}{15}a^7 \underbrace{\int_0^\infty x^6 e^{-x}dx}_{720=6!} =$$

$$= \frac{1}{\pi a}\frac{1}{64a^4}\frac{16\pi}{15}a^7 \cdot 720 = \frac{720\,a^2}{4\cdot 15} = 12a^2.$$

P4.14 The probability to be in interval $(r, r+dr)$ is

$$dP = p(r)\,dr = |\psi_{100}(r)|^2 r^2 dr \cdot 4\pi = \frac{4\pi}{\pi a^3} e^{-2r/a} r^2 dr$$

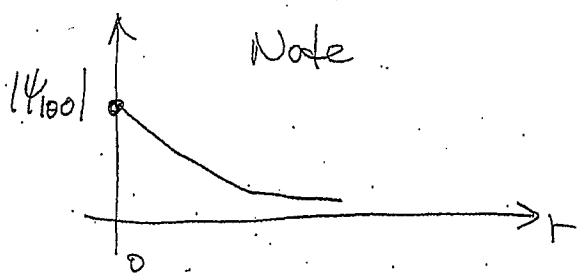
$\Rightarrow$ probability density $p(r) = \frac{4}{a^3} e^{-\frac{2r}{a}} r^2$

Most probable r = r that maximizes $p(r)$.

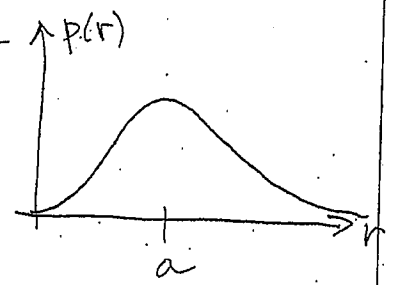
$$\frac{dp}{dr} = \left(\frac{4}{a^3}\right)\left(2r e^{-2r/a} - r^2\left(\frac{2}{a}\right)e^{-2r/a}\right) = 0$$

$$\Rightarrow 2r e^{-2r/a}\left(1 - \frac{r}{a}\right) = 0 \Rightarrow \boxed{r = a}$$

Note



but



$\boxed{4.15}$ Energy of a nth hydrogen atom state

$$E_n = \frac{E_1}{n^2}$$ ← Note this result does not depend on l or m!

$$\psi(r,0) = \frac{(\psi_{211} + \psi_{21-1})}{\sqrt{2}}$$

(a) $$\psi(r,t) = \frac{\psi_{211}\, e^{-iE_2 t} + \psi_{21-1}\, e^{-iE_2 t}}{\sqrt{2}} = e^{-iE_2 t}\,\frac{[\psi_{211} + \psi_{21-1}]}{\sqrt{2}}$$

$n=2,\ l=1;\ m=\pm 1$

$\boxed{\text{Eqn } 4.87}$ ← $L_0^5 = 5! = (-)^5 \left(\frac{d}{dx}\right)^5 L_5(x)$

$$\psi_{nlm} = \sqrt{\left(\frac{2}{2a}\right)^3 \frac{(3-1-1)!}{2\times 2\,[3!]^3}}\; e^{-r/2a} \left(\frac{2r}{2a}\right)^1 \left[\, L_0^5 \,\right] Y_1^m(\theta,\phi)$$

$$= \sqrt{\frac{1}{4(3!)^3}} \left(\frac{r e^{-r/2a}}{a}\right) L_0^5(r)\, Y_2^m(\theta,\phi)$$; where $n=2$ & $l=1$ have been used!

$$\frac{\psi_{211} + \psi_{21-1}}{\sqrt{2}} = \frac{1}{\sqrt{2}} \sqrt{\frac{1}{4(3!)^3}} \left(\frac{r e^{-r/2a}}{a}\right) L_0^5(r) \left[Y_1^1(\theta,\phi) + Y_1^{-1}(\theta,\phi)\right]$$

where $\rightarrow L_0^5 = 5!$

$$Y_1^{\pm 1} = \mp \sqrt{\frac{3}{8\pi}}\, \sin\theta\, e^{\pm i\phi}$$ } plugging it in

$$\Rightarrow \psi(r,t) = \frac{-i}{\sqrt{\pi a^2}}\, \frac{r e^{-r/2a}}{4a^2} \sin\theta \sin\phi\, e^{-iE_2 t/\hbar}$$

b

$$\langle V \rangle = \int d^3r \; \psi^*(r,t) \; V \; \psi(r,t)$$

$$= \int |\psi|^2 V \, r^2 dr \, \sin\theta \, d\theta \, d\phi$$

$$= \frac{1}{(2\pi a)\,16a^4} \int r^2 e^{-r/a} \left(\frac{-e^2}{4\pi\epsilon_0} \frac{1}{r}\right) \sin^3\theta \, \sin^2\phi \, r^2 dr \, d\phi \, d\theta$$

$$= \frac{1}{(2\pi a)\,16a^4} \left(\frac{-e^2}{4\pi\epsilon_0}\right) \left[a^4 \int_0^\infty \left(\frac{r}{a}\right)^3 e^{-r/a} d\frac{r}{a}\right] \left[\int_0^{2\pi} \sin^2\phi \, d\phi\right] \left[\int \sin^3\theta \, d\theta\right]$$

$$= \frac{1}{2\pi a \, 16a^4} \left(\frac{-e^2}{4\pi\epsilon_0}\right) \left[3! \, a^4\right] \left[\frac{2\pi}{2}\right] \left[-\underbrace{\int_0^\pi (1-\cos^2\theta) \, d\cos\theta}_{\left(\frac{4}{3}\right)}\right]$$

Note $\rightarrow$ $a = \dfrac{4\pi\epsilon_0 \hbar^2}{me^2}$ $\Rightarrow$ $\dfrac{e^2}{4\pi\epsilon_0} = \dfrac{\hbar^2}{ma}$

$$= -\frac{\hbar^2}{4ma^2}$$

$$= +\frac{E_1}{2}$$

$$= -6.8 \text{ eV}.$$

P4.11 (a) $R_{20}(r) = \frac{c_0}{2a}\left(1-\frac{r}{2a}\right)e^{-r/2a}$ (2)

Normalization: $1 = \int_0^\infty R_{20}^2(r)\, r^2 dr = \left(\begin{matrix} r = 2ax \\ dr = 2a\,dx \end{matrix}\right) =$

$$= \left(\frac{c_0}{2a}\right)^2 (2a)^3 \underbrace{\int_0^\infty (1-x)^2 e^{-2x} x^2 dx}_{1/4} = c_0^2 \cdot \frac{a}{2} \Rightarrow c_0 = \sqrt{\frac{2}{a}}$$

$$\Rightarrow R_{20}(r) = \sqrt{\frac{2}{a}}\,\frac{1}{2a}\left(1-\frac{r}{2a}\right)e^{-r/2a}$$

$$\psi_{200}(r) = R_{20}(r)\underbrace{Y_0^0(\theta,\varphi)}_{\frac{1}{\sqrt{4\pi}}} = \frac{1}{\sqrt{4\pi}}\,\frac{1}{\sqrt{2a^3}}\left(1-\frac{r}{2a}\right)e^{-r/2a}$$

$$= \frac{1}{\sqrt{2\pi a}}\,\frac{1}{2a}\left(1-\frac{r}{2a}\right)e^{-r/2a}$$

(b) $R_{21}(r) = \frac{c_0}{4a^2}\, r e^{-r/2a}$

$$1 = \left(\frac{c_0}{4a^2}\right)^2 \int_0^\infty r^2 e^{-r/a}\, r^2 dr = \left(\frac{c_0}{4a^2}\right)^2 a^5 \underbrace{\int_0^\infty x^4 e^{-x} dx}_{24}$$

$(r = a\cdot x)$

$$\Rightarrow \frac{c_0^2}{16a^4}\cdot 24a^5 = 1 \Rightarrow c_0^2 = \frac{2}{3a} \Rightarrow c_0 = \sqrt{\frac{2}{3a}}$$

$$\psi_{21\pm1}(r) = R_{21}(r)\, Y_1^{\pm1}(\theta,\varphi) = \sqrt{\frac{2}{3a}}\,\frac{1}{4a^2}\, r e^{-r/2a}$$

$$\left(\mp\sqrt{\frac{3}{8\pi}}\sin\theta\, e^{\pm i\varphi}\right) = \mp\frac{1}{\sqrt{\pi a}}\,\frac{1}{8a^2}\, r e^{-r/2a}\sin\theta\, e^{\pm i\varphi}$$

$$\psi_{210}(r) = R_{21}(r)\, Y_1^0(\theta,\varphi) = \sqrt{\frac{2}{3a}}\,\frac{1}{4a^2}\, r e^{-r/2a}\sqrt{\frac{3}{4\pi}}\cos\theta =$$

$$= \frac{1}{\sqrt{2\pi a}}\,\frac{1}{4a^2}\, r e^{-r/2a}\cos\theta.$$

$\boxed{P4.13}$ $\psi_{100} = \frac{1}{\sqrt{\pi a^3}} e^{-r/a}$ (3)

(a) $\langle r^n \rangle = \int \psi_{100}^* r^n \psi_{100} r^2 dr \underbrace{\sin\theta d\theta d\varphi}_{4\pi \text{ (full spherical angle)}}$

$$\Rightarrow \langle r \rangle = \frac{4\pi}{\pi a^3} \int_0^\infty r e^{-\frac{2r}{a}} r^2 dr = \frac{4}{a^3} a^4 \underbrace{\int_0^\infty x^3 e^{-2x} dx}_{\frac{3}{8}} = \frac{3a}{2}$$

$(r = ax)$

$$\langle r^2 \rangle = \frac{4\pi}{\pi a^3} \int_0^\infty r^4 e^{-2r/a} dr = \frac{4}{a^3} a^5 \underbrace{\int x^4 e^{-2x} dx}_{\frac{3}{4}} = 3a^2.$$

(b) $x = r \sin\theta \cos\varphi$

$\Rightarrow \langle x \rangle = 0$ in spherically symmetric $|\psi_{100}|^2$.

$\langle x^2 \rangle = \frac{1}{3} \langle r^2 \rangle$ by symmetry $\Rightarrow \langle x^2 \rangle = a^2$.

(c) $n = 2, \ell = 1, m = 1 \Rightarrow \psi_{211} = -\frac{1}{\sqrt{\pi a}} \frac{1}{8a^2} r e^{-r/2a} \sin\theta e^{i\varphi}$

(from P4.11b)

$|\psi_{211}|^2 = \frac{1}{\pi a} \frac{1}{(8a^2)^2} r^2 e^{-r/a} \sin^2\theta$ is <u>not</u> spherically symmetric!

$\Rightarrow$ expect finite $\langle x^2 \rangle$.

$$\langle x^2 \rangle = \int (r \sin\theta \cos\varphi)^2 |\psi_{211}|^2 r^2 dr \sin\theta d\theta d\varphi$$

$$= \frac{1}{\pi a} \frac{1}{(8a^2)^2} \int_0^\infty r^6 e^{-r/a} dr \underbrace{\int_0^\pi \sin^5\theta d\theta}_{-\int_0^\pi \sin^4\theta d(\cos\theta) =} \underbrace{\int_0^{2\pi} \cos^2\varphi d\varphi}_{\pi}$$

$$= -\int_0^\pi (1-\cos^2\theta)^2 d(\cos\theta) = \int_{-1}^1 (1-z^2)^2 dz = \frac{16}{15}$$